

Presentation Flow — 9-Pillar Evaluation Matrix

Team: Git with It · **Team ID:** 261 **PS 26153 (NTRO)** — AI based Network Attack Forecasting from Network Traffic Data

Ye document ek hi cheez ke liye hai: **har pillar deliberately hit ho**, ittefaq se nahi. Judges 9 alag cheezein score karte hain — agar aap sirf demo achha dikha doge to 3 pillar mil jaayenge aur 6 chhoot jaayenge.

Pehle ek nazar — kaunsa pillar kahan milta hai

#	Pillar	Kahan earn hota hai	Kaun bolta hai
1	Understanding of Problem Statement	Opening, 0:00–1:30	Suyash
2	Innovation & Novelty	"Cut the traffic off" moment, 1:30–3:00	Suyash
3	Technical Feasibility	Architecture + benchmark, 3:00–4:15 & 7:00–8:00	Suyash + Shivam
4	Impact & Scalability	Closing block, 8:00–9:00	Yuvan
5	Prototype / PoC	Live dashboard, 4:15–7:00	Tanay
6	Technology Competency	Stack + debugging stories, Q&A	Sab, apne area mein
7	Implementation & Live Demo	Play + upload, 4:15–7:00	Tanay + Yuvan
8	Presentation & Pitch Quality	Poore time	Suyash leads
9	Teamwork & Role Distribution	Opening 20 sec + Q&A handoffs	Sab

Sabse zyada chhootne wale pillar: 4, 6, 9. Teams demo mein kho jaate hain aur impact, competency aur teamwork bina bole nikal jaate hain. Neeche har ek ke liye jagah fix ki hai.

Minute-by-minute flow (10 minute + Q&A)

Time	Kya	Pillar	Kaun
0:00–0:20	Team intro, role division	9	Suyash
0:20–1:30	Problem: detection vs forecasting	1	Suyash
1:30–3:00	Solution + kya naya hai	2	Suyash
3:00–4:15	Architecture + tech stack	3, 6	Suyash
4:15–5:45	Live demo: replay, kill chain, explain	5, 7	Tanay
5:45–6:30	Topology + upload	5, 7	Tanay → Yuvan
6:30–7:00	"Hamesha 100% kyun?" — constructed sample	1, 3	Yuvan
7:00–8:00	Benchmarks + teen honest findings	3	Shivam
8:00–9:00	Impact, scale, deployment	4	Yuvan
9:00–9:30	Close: chaar line	8	Suyash
Q&A	Handoff by area	6, 9	Sab

Pillar 1 — Understanding of Problem Statement

Kahan: 0:20–1:30, aur phir 6:30 pe dobara (jab dataset ki limitation batate ho).

Bolna:

"PS kehta hai *forecast, detect* nahi. Ye farak poora project decide karta hai.

Aaj ke IDS — Suricata, Zeek — signature aur rule pe chalte hain. Wo batate hain **kya ho chuka**. Analyst ko alert tab milta hai jab attack ho chuka hota hai.

Humein jo chahiye tha wo alag sawaal hai: **ye host agle 10 minute mein kahan hoga?** Uske liye traffic classify karna kaafi nahi — network ki state kaise badalti hai, wo seekhna padta hai."

Ye bhi bol dena (understanding yahin dikhti hai):

"PS ne flow-level aur packet-level dono maange the — humne 39 flow features aur 23 packet features banaye. MITRE ATT&CK stage mapping maangi thi — 5 stages. Explainability maangi thi, black box nahi — teen channel diye. Aur fully offline, no cloud API — poora system CPU pe chalta hai."

Trap: PS ko "network security ka problem" bol ke general baat mat karna. Uske exact words uthao — forecast, ATT&CK, explainable, offline.

Pillar 2 — Innovation & Novelty

Kahan: 1:30–3:00. Ye pitch ka sabse important 90 second hai.

Bolna — chaar step mein:

- "1. Traffic ko 60-second windows mein todte hain. Har window ek host ka 39-feature snapshot hai.
2. Model transitions seekhta hai — is state ke baad aam taur pe kaunsa state aata hai.
3. Phir **hum traffic dikhana band kar dete hain**. Model ko kuch nahi milta.
4. Aur wo agle 10 state **khud banata hai**. Har imagined state pe hum poochhte hain — risk kya hai, ATT&CK stage kya hai."

Step 3 pe rukna. Wahi novelty hai. Classifier ye kar hi nahi sakta.

Novelty ka proof (ye zaroor bolna):

"Aur ye claim humne testable banaya hai. Humare test suite mein ek assertion hai: agar sirf imagination loss se backprop karo, to **encoder ka gradient zero aana chahiye** — kyunki imagination mein encoder use hi nahi hota. Agar wo test fail ho, matlab hum cheat kar rahe hain. Wo pass hota hai."

Trap: "world model" bol ke aage mat badhna. Judge ko ye phrase pata ho sakta hai. **"Traffic band kar dete hain"** wala step bolo — wo yaad rehta hai.

Pillar 3 — Technical Feasibility

Kahan: 3:00–4:15 (architecture) aur 7:00–8:00 (benchmarks).

Architecture ek line mein:

"Ingest → 39+23 features → 60-second state cells → encoder+GRU → prior rollout → teen heads → dashboard. Slide pe poora diagram hai."

Feasibility ke paanch point:

Technical	1,38,398 parameters, CPU-only. GPU nahi chahiye.
Financial	Zero licence cost, sab open source
Operational	NetFlow/IPFIX jo enterprise already export karte hain
Legal	Data network se bahar jaata hi nahi
Social	Final decision analyst ka, model sirf ranking deta hai

Benchmarks (Shivam):

"Split time se kiya, randomly nahi — har capture ke past pe train, future pe test. Baselines ko bilkul wahi features, wahi scaler, wahi labels diye.

Stage forecast +3 windows pe hum 0.647 dete hain, 'maan lo kuch nahi badlega' wala baseline 0.478. 10 mein se 9 horizons pe hum aage hain."

Trap: Detection F1 ko badha-chadha ke mat bolna. Wo 0.979 vs 0.977 — tie hai. Khud bol do, judge ke poochhne se pehle. Neeche Finding 1 mein hai.

Pillar 4 — Impact & Structural Scalability

Kahan: 8:00–9:00. **Ye sabse zyada chhootne wala pillar hai** — demo ke baad log seedha close kar dete hain.

Bolna (Yuvan):

"Kaun use karega — SOC analysts ko alert flood ki jagah ranked queue milta hai. Critical infrastructure: power, banking, telecom, transport. CERT-In jaise national response teams.

Scale: inference batched hai — ek capture ke saare hosts ek pass mein. 2,836 hosts wale capture pe first load ke baad sub-second triage hai.

Deployment: naya hardware nahi chahiye. Jo NetFlow already export ho raha hai wahi input hai. Ek Python process, ek checkpoint file. Air-gapped network mein bhi chalega kyunki kuch bahar jaata hi nahi."

Trap: "lakhs of hosts pe scale karega" mat bolna — humne test nahi kiya. Jo test kiya hai wo number bolo: **2,836 hosts**. Aur aage ke liye kaho "distributed karna seedha hai, par humne measure nahi kiya."

Pillar 5 — Prototype / Proof of Concept

Kahan: 4:15–7:00. Dashboard khula rakhna, slides pe wapas mat jaana.

Ye pillar **prototype ka asli hona** score karta hai — kitna complete hai, kitna kaam karta hai. Demo ka order Pillar 7 mein hai.

Bolne ki line:

"Ye mock nahi hai. Real CTU-13 data, 2,58,229 network states, 13 captures. Model abhi is laptop pe chal raha hai — koi cloud call nahi."

Pillar 6 — Technology Competency

Kahan: Q&A mein zyada, aur architecture ke waqt thoda.

Ye pillar tab milta hai jab aap **kyun** ka jawab de paate ho, sirf **kya** ka nahi. Har member apne area ke 2-3 numbers yaad rakhe.

Sabse strong competency signal — debugging stories:

"Pandas 3.0 mein timestamps default `datetime64[us]` hain. Humara code nanoseconds maan raha tha, to saare timestamps 1000 guna galat aa rahe the. `.dt.as_unit('ns')` se fix kiya."

"Model selection humne BCE loss pe nahi, validation average precision pe kiya. Data 0.8% positive hai — wahan loss galat model chun leta hai."

"Scapy ka poora stack slow hai, isliye `RawPcapReader` + `struct` use kiya — packet objects banaye bina headers parse karte hain. 4 lakh packets 12 second mein."

Trap: Agar kisi member se uske area ka sawaal aaye aur jawab na aaye — bluff mat karna. Bolna: "*Wo hissa maine implement nahi kiya, Suyash ne kiya.*" Galat jawab pakda jaata hai, "maine wo part nahi kiya" nahi.

Pillar 7 — Implementation & Live Demo

Kahan: 4:15–7:00. Exact click order:

1. Host list (10 sec) Scenario 1. Sabse upar `147.32.84.165` — 100%. Baaki 0%.

"Model ne wahi machine pakdi jisme asli malware tha."

2. Play dabao (60 sec) — sabse strong moment

"Divider ke baayein model traffic dekh raha hai. Daayein kuch nahi — har point wo khud bana raha hai."

Play ke dauraan **teen cheezon pe ungli rakhna:** - playhead baayein se daayein safar karta hai - neeche **kill chain** ka NOW/PREDICTED card live badalta hai - **~12 second pe ek tone bajega** — wahi stage change hai. Us beat pe kaho: *"Ye sun rahe ho? model ne abhi stage change detect kiya."*

Normal speed pe poora capture ~1 minute. Time kam ho to **Fast** (~20 sec) ya 15 second baad Pause.

3. Pehle hi bol dena (warna judge poochhega):

"Probability 100% pe flat hai. Jo badal raha hai wo **line ka rang** hai — wahi ATT&CK stage hai."

4. Explainability (30 sec) Feature attribution + temporal attention.

"Model ke paas 120 minute ka past tha. Barabar baantta to har minute ko 0.8% milta. Usne 25% attention sirf 5 minute pe daala — sabse zyada 23 minute pehle wale pe, average se 7.7 guna."

5. Topology tab (30 sec) 3D scene. Scroll karke zoom, ek node pe click.

"Node ka rang humara model ka conclusion hai, edge ka rang dataset ka label. Isliye hara node laal edges ke saath dikhe to wo host humara model **miss** kar gaya. Humne chhupaya nahi."

6. Benchmark tab (20 sec) Horizon chart pe cursor ghumao — har point ke exact numbers.

7. Upload — "hamesha 100% kyun?" (30 sec) ← Pillar 1 + 3 dobara `data/samples/host-becomes-infected.csv` upload karo.

"Ye file humne **banayi** hai — dataset mein aisa host tha hi nahi. Ek saaf workstation ka traffic, phir usi IP pe botnet traffic. Ab dekhiye — 38 window flat, phir risk chadhta hai. Aur surprise channel **theek us window pe** $+6\sigma$ spike karta hai jab traffic badalta hai, risk head 10 window baad commit karta hai. Isiliye humne dono channel rakhe hain."

Trap: File constructed hai — **pehle hi bolna**. Baad mein pata chalna bahut bura hota hai.

Pillar 8 — Presentation & Pitch Quality

Ye pillar alag block mein nahi milta — poore time milta hai.

Chaar cheezein jo score badalti hain:

1. **Ek waqt mein ek banda bolta hai.** Beech mein add-on mat karna. Handoff saaf ho: "*Iska detail Shivam batayega.*"
2. **Slide padhna nahi hai.** Slide pe diagram hai; aap uska matlab bolo.
3. **Numbers exact bolna.** "kaafi accha" ki jagah "0.647 vs 0.478".
4. **Limitations khud bolna.** Judge ke poochhne se pehle bola hua limitation confidence dikhata hai; poochhne ke baad bola hua bahana lagta hai.

Closing chaar line (Suyash, 9:00–9:30):

"Ek — hum classify nahi karte, network ki state aage chalate hain. Do — detection mein hum baseline ke barabar hain, aur forecasting mein 10 mein se 9 horizons pe aage. Teen — har alert ke saath teen explanation hain, black box nahi. Chaar — sab kuch offline chalta hai, is laptop pe, abhi."

Pillar 9 — Teamwork & Role Distribution

Kahan: Opening ke 20 second, aur phir har Q&A handoff pe.

Opening line (Suyash):

"Main Suyash, team lead — maine world model aur architecture pe kaam kiya. Team ke baaki paanch: data pipeline, evaluation, dashboard, backend, aur MITRE mapping. Har area ka owner yahan hai — jo detail chahiye poochh lijiye."

Role map:

Member	Area	Q&A mein kya own karta hai
Suyash Shukla	Team Lead · World Model	RSSM, imagination rollout, architecture
Tanmay Kumar Sinha	Data Pipeline & Features	CTU-13 ingest, 39+23 features, windows
Shivam Kumar Tripathi	Evaluation & Benchmarks	splits, baselines, saare numbers
Tanay	Dashboard & Visualisation	replay, topology, kill chain
Yuvan Laxmanan	Backend & Deployment	FastAPI, upload, offline deployment
Vaishnavi Tripathi	MITRE Mapping & Explainability	5 stages, IG, attention

Trap: Agar sirf ek banda poore time bole to Pillar 9 zero ho jaata hai — chahe demo kitna bhi accha ho.
Har member ko kam se kam ek baar bolna hai. Jo blocks upar assign kiye hain, wahi kaafi hain.

Teen honest findings — Pillar 3 aur 8 dono ke liye

Ye tab bolna jab judge deep jaaye, ya 7:00–8:00 wale block mein.

Finding 1 — Humne apne hi code mein cheating pakdi

"Ek metadata flag — `has_packet_features` — perfect label proxy ban gaya tha, kyunki humare paas PCAP sirf botnet traffic ka tha. F1 0.23 se 0.98 ho gaya jab hataya. Ye humne khud audit mein pakda."

Finding 2 — Dataset mein pre-infection baseline hai hi nahi

"2,58,229 windows check kiye. Infected hosts ka **ek bhi clean window nahi**. Isiliye probabilities calibrated nahi hain — 1,530 hosts mein har ek ya ~100% hai ya <5%. Aur isiliye humne binary detection ko main task nahi banaya."

Finding 3 — Do channels ulte fail hote hain

"Supervised head sustained activity pe kaam karta hai, surprise channel short bursts pe. Upload wale demo mein ye dikha — surprise us window pe fire hua, risk head 10 window baad. Dono ulte hain, isiliye dono lagaye. Milake 30 mein se 28 host, 4.9% false alarms pe."

Numbers — ek jagah

Network states	2,58,229
Captures	13 (CTU-13, CC BY)
Model	1,38,398 parameters, CPU-only
Features	39 flow + 23 packet, 60-second windows
Detection F1	0.979 (baseline 0.977 — tie)
False positive rate	0.0009
Stage forecast +3	0.647 (baseline 0.478)
Horizons jeete	9 of 10
Hosts caught	28 of 30, 4.9% false alarms
Unseen families	F1 0.874, ROC-AUC 0.982
PCAP throughput	4 lakh packets → 18,271 flows → 12 sec
Training	12 epochs, early stop (best epoch 10)

Aakhri check — nikalne se pehle

- [] Server chal raha hai, `localhost:8000` khula hai
- [] Scenario 1 pre-loaded, host `147.32.84.165` selected
- [] Sound toggle  on hai (ya jaan-boojh ke off, agar room audio kharab ho)
- [] `data/samples/host-becomes-infected.csv` ka path pata hai
- [] Deck ki PDF bhi khuli hai (backup, agar projector PPT na le)
- [] Har member ko apne 2-3 numbers yaad hain